

Word-Level Plagiarism Checker for Short Documents

A word-level similarity detection system for comparing short documents and identifying potentially copied content.

TEAM MEMBERS

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UNDER THE GUIDANCE OF

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Why existing checkers fall short

Traditional plagiarism tools rely on exact-phrase matching and external services — they miss the ways real students actually rephrase text.

THE CHALLENGE

-  **Reordered at the word level**
Copied text with word order shuffled to evade detection
-  **Reworded Content**
Similar content expressed using different words
-  **Partially copied from multiple sources**
Composite plagiarism assembled from several documents
-  **Simple Matching Limitations**
Basic matching may miss reordered or reworded content
-  **Limited Comparison Approaches**
Basic matching may miss reordered or reworded content



OUR PROBLEM GOAL

Build an efficient checker for short documents that:

- ✓ Detects exact & near-exact word-level similarities
- ✓ Identifies reordered and partially matching content
- ✓ Detects basic reworded content
- ✓ Locates suspicious text segments
- ✓ Identifies the probable source document
- ✓ Produces a clear, readable similarity report



Key DSA Challenge: Achieve accurate matching while maintaining efficient time and space complexity — using only custom-built data structures and string-processing techniques.

Existing systems vs. our proposed system

ASPECT	EXISTING SYSTEMS	OUR PROPOSED SYSTEM
Core Approach	✗ External plagiarism databases & APIs	✓ Locally maintained reference corpus
Detection Focus	✗ Exact phrase / semantic similarity	✓ Word-level exact, reordered & partial similarity
Data Structures	✗ Different algorithms and software tools	✓ DSA-based data structures and algorithms
Connectivity	✗ Requires internet / API access	✓ Works offline with a local corpus
Transparency	✗ Limited insight into matching logic	✓ Transparent similarity results & source attribution
Target Scale	✗ Large-scale commercial environments	✓ Optimized for short documents & interactive analysis
Control	✗ Limited control over internal logic	✓ DSA concepts used in the matching process

Key Differentiator: *A lightweight similarity checker focused on short documents, word-level comparison, and transparent matching results.*

Proposed system architecture

A query document flows through a structured, multi-stage pipeline and is compared against a reference corpus.



Core Principle: Efficient word-level comparison using DSA-based matching and comparison techniques.

Key features & the data structures behind them

KEY FEATURES

- 

Word-Level Detection
Identifies matching words & phrases across documents
- 

Reordered & Reworded Detection
Identifies similar content with changes in word order or wording
- 

Partial Similarity Detection
Identifies copied segments within a larger document
- 

Source Attribution
Shows which reference document contains matching content
- 

Similarity Score & Report
Calculates and presents document-level similarity clearly

DSA COMPONENTS

- 

Hash Table
Fast word lookup & indexing
- 

Sliding Window
Detect reordered word sequences
- 

Arrays / Linked Structures
Store tokens & document data
- 

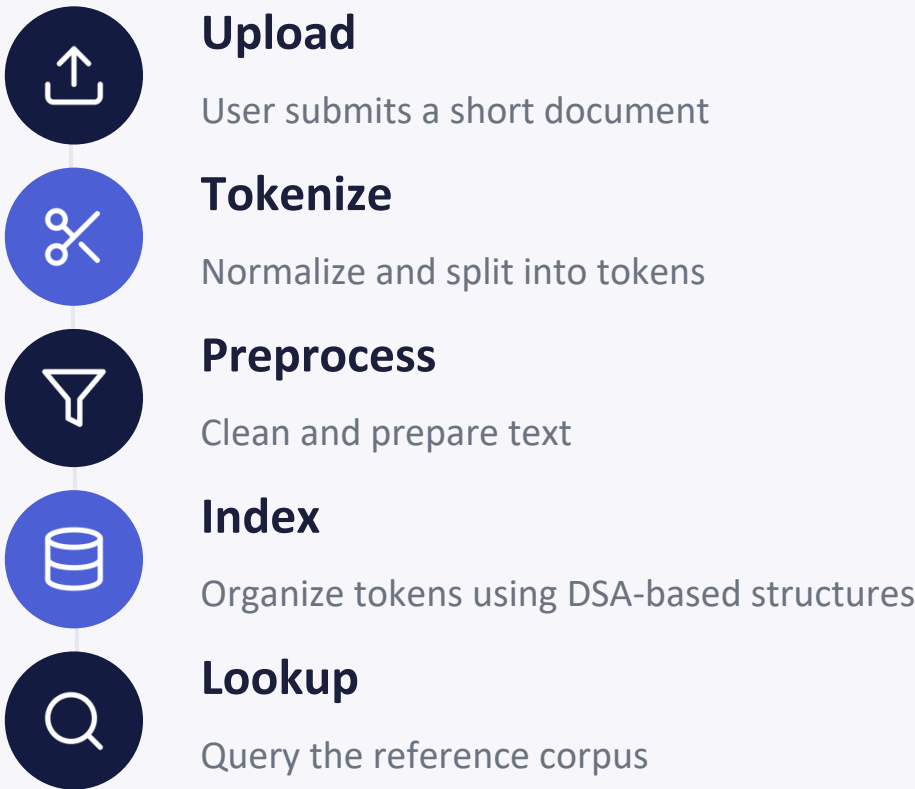
String Matching
Detect matching sequences
- 

Frequency Maps
Compare word occurrence patterns across documents

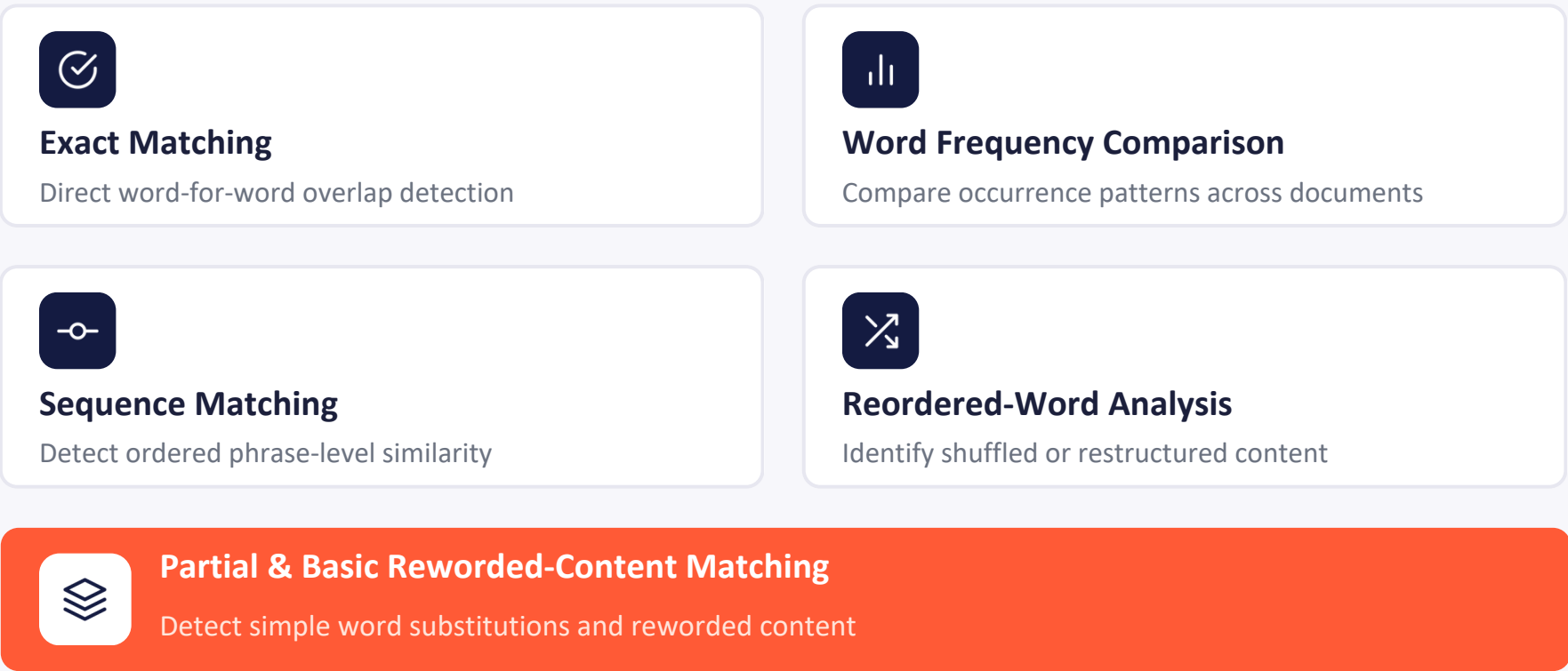
Goal: *Balance detection accuracy with efficient execution using DSA-based matching techniques.*

System architecture & working

SYSTEM FLOW



SIMILARITY ENGINE — INTERNAL COMPONENTS



CONCEPTUAL OUTPUT REPORT

XX%

Overall Similarity

Source 1 — XX% | Source 2 — XX%

Source Attribution

Highlighted Segments

Suspicious text flagged with source reference

Outcome & project results

EXPECTED / ACHIEVED OUTCOMES

-  **Document Comparison**
Successfully compares short documents against a reference corpus
-  **Overlap Detection**
Detects exact word/phrase overlap and partial copied segments
-  **Reordered & Reworded Matching**
Identifies similar content with changes in word order or basic wording
-  **Source Attribution**
Attributes suspicious content to probable source documents
-  **Similarity Report**
Generates a clear, human-readable similarity report

DSA LEARNING OUTCOME

-  **Hashing**
Fast indexing & lookup
-  **String Processing**
Tokenization & normalization
-  **Searching Algorithms**
Efficient pattern detection
-  **Custom Structures**
DSA-based structures for efficient comparison
-  **Algorithmic Efficiency**
Optimized time & space complexity

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The project transforms theoretical DSA concepts into a real-world text-analysis application.

Future Scope

- 1

Corpus Expansion

Expand the reference corpus for broader coverage
- 2

Multilingual Support

Support multilingual plagiarism detection
- 3

Advanced Semantic Similarity

Improve detection of heavily reworded content
- 4

Web Deployment

Develop a browser / web-based interface
- 5

ML-Assisted Classification

Machine-learning-aided plagiarism classification

Conclusion

Word-Level Plagiarism Checker provides a practical approach to detecting copied and lightly modified content in short documents.

IT COMBINES



to create a **transparent and efficient** plagiarism-analysis system.

“Detect Similarity. Trace the Source. Protect Originality.”

Thank You | Questions?

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